# Flood Image Classification

Baseline Results Summary (Seed 42) | CVFD Dataset | Tesla T4 GPU

## Cross-Model Comparison

| Model | Params | Test Acc | F1 | Precision | Recall | ROC AUC | PR AUC |
|---|---|---|---|---|---|---|---|
| **ViT-Small DINOv3** | **21.6M** | **98.98%** | **0.9892** | **0.9874** | **0.9909** | **0.9995** | **0.9994** |
| Swin-Base | 86.7M | 98.64% | 0.9855 | 0.9838 | 0.9873 | 0.9980 | 0.9975 |
| ConvNeXt V2-Tiny | 27.9M | 98.14% | 0.9800 | 0.9853 | 0.9746 | 0.9985 | 0.9984 |
| ConvNeXt-Tiny | 27.8M | 97.80% | 0.9767 | 0.9663 | 0.9873 | 0.9988 | 0.9986 |
| ResNet-50 | 23.5M | 95.17% | 0.9492 | 0.9335 | 0.9656 | 0.9930 | 0.9921 |
| EfficientNetV2-S | 20.2M | 91.44% | 0.9096 | 0.8991 | 0.9203 | 0.9777 | 0.9749 |
| MobileNetV3-Large | 4.2M | 87.46% | 0.8669 | 0.8607 | 0.8732 | 0.9512 | 0.9482 |

* Green row = best performing model. All models trained with identical pipeline: two-stage fine-tuning, pretrained weights, full augmentation, LR warmup.

## Dataset & Training Configuration

| Parameter | Value |
|---|---|
| Dataset | CVFD (Close-View Flood Dataset) |
| Train / Val / Test split | 5,496 / 1,177 / 1,180 images |
| Class balance (Flooded / Non-Flooded) | 2,573 / 2,923 train \| ~47% / 53% |
| Class imbalance | 6.4% — mild, no correction applied |
| Batch size | 32 |
| Stage 1 (head-only) | 5 epochs, LR warmup |
| Stage 2 (full fine-tune) | up to 20 epochs, cosine LR decay, early stopping |
| Hardware | Tesla T4 GPU \| PyTorch 2.10.0+cu128 |
| Seed | 42 (single seed — multi-seed runs pending) |

## Per-Class Classification Reports

**ViT-Small DINOv3** (vit_small_patch16_dinov3.lvd1689m)

| Class | Precision | Recall | F1 | Support |
|---|---|---|---|---|
| Non-Flooded | 0.9920 | 0.9889 | 0.9904 | 628 |
| Flooded | 0.9874 | 0.9909 | 0.9892 | 552 |
| Accuracy | | | 0.9898 | 1180 |

| | | | | |
|---|---|---|---|---|
| Macro Avg | 0.9897 | 0.9899 | 0.9898 | 1180 |

**Swin-Base** (swin_base_patch4_window7_224)

| Class | Precision | Recall | F1 | Support |
|---|---|---|---|---|
| Non-Flooded | 0.9888 | 0.9857 | 0.9872 | 628 |
| Flooded | 0.9838 | 0.9873 | 0.9855 | 552 |
| Accuracy | | | 0.9864 | 1180 |
| Macro Avg | 0.9863 | 0.9865 | 0.9864 | 1180 |

**ConvNeXt V2-Tiny** (convnextv2_tiny.fcmae_ft_in22k_in1k)

| Class | Precision | Recall | F1 | Support |
|---|---|---|---|---|
| Non-Flooded | 0.9779 | 0.9873 | 0.9826 | 628 |
| Flooded | 0.9853 | 0.9746 | 0.9800 | 552 |
| Accuracy | | | 0.9814 | 1180 |
| Macro Avg | 0.9816 | 0.9809 | 0.9813 | 1180 |

**ConvNeXt-Tiny** (convnext_tiny.in12k_ft_in1k)

| Class | Precision | Recall | F1 | Support |
|---|---|---|---|---|
| Non-Flooded | 0.9886 | 0.9697 | 0.9791 | 628 |
| Flooded | 0.9663 | 0.9873 | 0.9767 | 552 |
| Accuracy | | | 0.9780 | 1180 |
| Macro Avg | 0.9775 | 0.9785 | 0.9779 | 1180 |

**ResNet-50** (resnet50.a1_in1k)

| Class | Precision | Recall | F1 | Support |
|---|---|---|---|---|
| Non-Flooded | 0.9688 | 0.9395 | 0.9539 | 628 |
| Flooded | 0.9335 | 0.9656 | 0.9492 | 552 |
| Accuracy | | | 0.9517 | 1180 |
| Macro Avg | 0.9511 | 0.9525 | 0.9516 | 1180 |

**EfficientNetV2-S** (tf_efficientnetv2_s.in21k_ft_in1k)

| Class | Precision | Recall | F1 | Support |
|---|---|---|---|---|
| Non-Flooded | 0.9285 | 0.9092 | 0.9187 | 628 |
| Flooded | 0.8991 | 0.9203 | 0.9096 | 552 |
| Accuracy | | | 0.9144 | 1180 |
| Macro Avg | 0.9138 | 0.9148 | 0.9142 | 1180 |

**MobileNetV3-Large** (mobilenetv3_large_100.ra_in1k)

| Class | Precision | Recall | F1 | Support |
|---|---|---|---|---|
| Non-Flooded | 0.8871 | 0.8758 | 0.8814 | 628 |
| Flooded | 0.8607 | 0.8732 | 0.8669 | 552 |
| Accuracy | | | 0.8746 | 1180 |
| Macro Avg | 0.8739 | 0.8745 | 0.8742 | 1180 |

Note: All results are from a single seed (42).